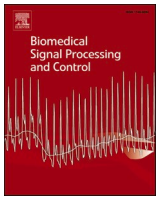
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Oral cancer detection using feature-level fusion and novel self-attention mechanisms

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ABSTRACT

The rising prevalence of oral and dental conditions, including issues like gum disease and oral cancer, presents a pressing global health challenge. The promptly identification of individuals at risk of developing these potentially life-threatening oral ailments, especially those linked to lip and oral cavity cancer. Early detection is crucial for effective intervention and containment of these diseases. Several studies in medical research have explored the use of convolutional neural network (CNN) models with pre-trained weights to identify oral and dental diseases. Previous research has primarily focused on amalgamating predictions from various models to boost accuracy, the integration of diverse model predictions presents challenges such as potential inconsistencies, conflicting outcomes, and heightened model complexity. However, utilizing a variety of model features can not only boost the robustness and accuracy of the detection process but also streamline the overall complexity of the model ensemble. The challenge lies in finding more effective methods to combine and leverage features extracted from deep learning models for superior predictive accuracy. To tackle this challenge, our research introduces an innovative approach that incorporates feature-level fusion techniques for oral cancer detection. We have developed a novel self-attention block as a foundational element of our approach. Additionally, we utilize transfer learning (TL) models for feature fusion, specifically EfficientNetB0 and EfficientNetB1, renowned for their effectiveness in image classification tasks. During the training process, we incorporate a self-attention block, a crucial innovation that facilitates the fusion of features extracted from these EfficientNet models. Our proposed methodology revolves around the concept of feature-level fusion. We employ a novel self-attention block and leverage transfer learning models to extract and combine relevant features. This fusion strategy enables us to collective strengths of these models, resulting in more accurate predictions than what can be achieved using each model in isolation. To evaluate the effectiveness of our approach, we conducted rigorous assessments using the well-established MOD dataset, which is a publicly available benchmark in the field of oral cancer detection. Our model achieved an impressive accuracy rate of 98.83% in these evaluations.

1. Introduction

Cancer is a malignant disease characterized by uncontrolled cell proliferation and invasion of surrounding tissues. Metastasis is the process through which cancer cells move to different areas of the body. Oral cancer is a rapidly spreading form of cancer. It initially manifests as non-malignant growths, red rashes, or mouth sores before developing into cancer [1]. This type of cancer typically originates in the mouth or its vicinity. It can affect the different part of mouth. An estimated 500,000 or more persons are diagnosed with oral cancer annually, making it the sixth most frequent type of cancer worldwide. Numerous variables have been linked to an increased risk of oral cancer, such as exposure to sunlight, alcohol and tobacco use, human papillomavirus (HPV) infection, poor dental hygiene, and a low diet of fruits and vegetables [2]. Symptoms of oral cancer can include a sore on the lips or gums that does not heal, a white or red patch on the mouth lining, difficulty swallowing or chewing and a lump or growth in the neck. Traditional methods for detecting oral cancer include visual inspection, oral cytology, and biopsy. Visual inspection involves examining the mouth and throat for abnormalities, while oral cytology involves collecting cells from the mouth and examining them under a microscope. Biopsy is the most definitive test for oral cancer and involves removing a small sample of tissue for examination under a microscope. Advanced medical imaging techniques, such as MRI, X-ray, and CT scans, can

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diagnose oral cancer earlier and more precisely than traditional methods such as visual inspection and biopsy, enabling more effective treatment [3]. Oral cancer is a prevalent malignancy with various risk factors. While traditional detection methods may have limitations, modern medical imaging techniques can diagnose oral cancer earlier and more accurately, ultimately improving the effectiveness of treatment.

Machine Learning (ML) algorithms are increasingly being used to interpret and evaluate medical images, including oral cancer images. This can help to reduce medical costs and improve the accuracy of diagnosis. ML algorithms can autonomously identify occult patterns in medical images, such as oral cancer images, obviating the need for extensive human intervention. In [4] employed optical devices to evaluate the imaging characteristics of oral cancer, various other oral diseases, and potentially malignant oral lesions. Mouth cancer screening is crucial for early diagnosis and treatment, which improves patient survival rates. Deep Learning (DL) [5] to automate the detection of oral cancer lesions, enabling early diagnosis, a critical need in low and middle-income countries with high prevalence. To facilitate the evaluation of two DL methods for cancer detection, they developed an annotation tool to create a well-curated oral lesion library. Study [6] developed a DL approach to detect abnormal oral squamous cells, a critical step in the early diagnosis of oral cancer, which can reduce mortality and treatment costs. ML and DL algorithms have shown great promise in medical imaging, but they also have drawbacks such as requiring large amounts of data, being computationally expensive, and being susceptible to overfitting. CNNs are well-suited to address these challenges, as they are able to learn complex spatial patterns in images from relatively small datasets and are computationally efficient. Image processing techniques significantly improve the visibility and quality of oral cancer images, facilitating early and accurate detection of cancerous lesions. This enhancement of image quality empowers clinicians to provide timely diagnosis and more effective treatment, ultimately resulting in better patient outcomes. Image processing techniques, such as image resizing and filtering, were used to enhance the quality and reduce noise in oral cancer images. Furthermore, Feature fusion is a ML technique that combines multiple features from different sources to improve the performance of a model. It has been shown to be effective for improving the performance of oral cancer detection models. Feature fusion [7] has the potential to help clinicians detect oral cancer at an earlier stage, which can lead to better patient outcomes. Medical image fusion aims to combine complementary information from multiple medical images to produce a fused image with enhanced quality and informativeness, facilitating accurate diagnosis and effective treatment.

TL [8–11], which utilizes pre-trained models from large-scale image classification tasks, has become a popular approach in medical imaging, especially for the identification of oral cancer. In [6,8] the specialized domain of head, neck, and oral cavity imaging, there has been significant research focused on classifying the mouth and lip regions using photographic images. Study [12] compared two simple CNN models to more advanced TL models for medical image classification. The simple CNN models achieved modest accuracy (75.81 % and 76.61 %), while TL with MobileNet, ResNet, and Inception V3 significantly improved accuracies (85.48 %, 91.13 %, and 89.52 %, respectively). This suggests that TL is a promising approach for future medical image classification research. In [13] showed that TL can improve the performance of DL models for oral cancer detection, and that the Inception-V3 algorithm outperforms other algorithms. Authors [14] demonstrated the effectiveness of TL for developing DL-based computer-aided diagnostic systems for oral cancer detection. They fine-tuned the ResNet50 pre-trained model on a small dataset of oral squamous cell carcinoma (OSCC) histopathology images and achieved an accuracy of 96.6 %, which is a significant improvement over the baseline model trained from scratch (85.2 %). Study [15] used TL to classify OSCC into four distinct classes based on Broder's histological grades. ResNet-50 achieved the highest accuracy at 92.15 % among four pre-trained CNN models, but the authors also proposed their own CNN model, which outperformed the TL approaches with an accuracy of 97.5 %. Proposed a feature fusion [16] approach for oral cancer detection that combines BOVW and Inception-V3 features, achieving an accuracy of 97 %. The concatenated features were fed to ML classifiers. The authors also proposed a novel improved BOVW technique. Study [17] proposed a deep multi-feature fusion residual network (DMFF-ResNet) model for oral cancer classification, which achieved a competitive accuracy of 93.28 % on a dataset of 90 patients with OSCC, outperforming state-of-the-art classifiers. The DMFF-ResNet model takes the output spectral features of the last three blocks of a 1-dimensional ResNet50 network as input and extracts more spatial representations and discriminative power through feature fusion. Authors [18] developed and evaluated two DL models, DenseNet121 and faster R-CNN, for the classification and detection of oral cancer lesions in clinical oral photographs. The DenseNet121 model achieved an accuracy of 99 %, while the faster R-CNN model achieved an accuracy of 76.67 %.In a recent study, [19] demonstrated that Sharpness Aware Minimization (SAM) outperforms stochastic gradient descent (SGD) in training a VGG16 CNN for oral exfoliative cytology classification, achieving higher accuracy (93.28 % vs. 89.12 %) and AUC (0.9328 vs. 0.9217).SAM is a promising optimizer for CNNs used in oral exfoliative cytology classification. It can help to improve classification accuracy and suppress overfitting.

Recently developed [20] a DL model for the identification and segmentation of oral epithelial dysplasia (OED) in whole-slide pathology images. Their model, based on the DeepLabv3+ architecture, achieved an overall accuracy of 93.3 % on a held-out test set of 44 whole-slide images. This study demonstrates the feasibility of using CNN-based DL models to improve the accuracy and efficiency of OED diagnosis. DeepLabv3+ outperformed UNet++ in OED segmentation and classification tasks. Proposed [21] a novel DL-based approach for oral cancer diagnosis using a metaheuristic approach. The approach achieved an accuracy of 98.5 %, outperforming other techniques in the literature. Authors [22] conducted a comparative analysis of six ML algorithms for the classification of oral cancer using histopathological images. The neural network outperformed the other models with an accuracy of 90.4 %. This suggests that neural networks are a promising tool for the screening of oral cancer.

One significant categorization problem in artificial intelligence is diagnosing oral cancer from oral and cavity images. Many studies [15,20,23] have demonstrated the efficacy of innovative model designs and modifications of existing ones through transfer learning, a crucial approach given the scarcity of available datasets. The stack ensemble technique captures the diversity among various base models, often resulting in enhanced predictive performance. Feature-level fusion is another approach that can boost the overall representation of the data, potentially leading to improved model generalization and performance. Unlike fusion methods primarily focused on enhancing predictions through model diversity, feature-level fusion prioritizes feature representation. For instance, FuseNet [24] combines multimodal input (visual and depth images) to identify in skin situations semantically. Intermediate feature mappings are integrated through elementwise addition before propagating to the visual encoder stream. RTFNet [25] employs a similar fusion technique for urban scene semantic segmentation. It enhances model performance by introducing an upception block with transposed convolution and batch normalization layers. The MFNet [26], encoder's feature maps influence the decoder layer, with the encoder's output being added at a matching level in the decoder.

Our study introduces an innovative approach that merges two TL models, specifically EfficientNetB0 and EfficientNetB1. This fusion incorporates an initial naïve module, setting it apart from earlier methodologies that primarily focused on proposing entirely new network architectures. The novel self-attention block plays a pivotal role in this approach by enhancing feature extraction capabilities while preserving existing feature relationships. The main contribution of this study as follows;

- The proposed methodology has developed an optimal feature fusion framework that efficiently merges features of two models based on EfficientNet. This framework has proven its efficacy in improving the accuracy of oral cancer detection in patients by utilizing oral and cavity images. Utilizing feature-level fusion presents a valuable method for refining feature representation, potentially leading to considerable enhancements in model performance and generalization.
- A novel self-attention block has been designed to improve features extracted by backbone models. This block utilizes a combination of dense layers, repeated vectors, and multiplication layers to augment features by integrating information from fused features across varying scales.
- In comparison to convolutional layers, the dense layer, repeated vector, and multiply layer approach offers increased adaptability and flexibility in capturing complex patterns and relationships within the feature space, leading to more accurate and effective feature enhancement.
- The proposed model underwent evaluation using the openly accessible MOD dataset, demonstrating superior performance compared to the TL model and previous techniques. Additionally, its robustness was confirmed through testing on a diverse multi-class oral and cavity MOD dataset. These results emphasize the model's exceptional capabilities and its potential for practical deployment in real-world scenarios.

2. Materials and methods

2.1. Mouth and oral disease (MOD) dataset

The original MOD [27] dataset comprises 517 samples. An expanded version of the MOD dataset, publicly accessible, includes 5143 image samples. The distribution of samples across each class is visualized in Fig. 1, while Table 1 summarizes the sample counts for each class.

2.2. Gamma correction

When captured in a color image acquisition system, a digital camera's color channels (R, G, and B) exhibit a non-linear response. Employing gamma correction [28] can significantly enhance the discernibility of image details, particularly in scenarios with low light or high contrast. In this study, we have applied the gamma correction technique with a 1.5 gamma value, which proves valuable in unveiling concealed elements within shadowed or overexposed regions and compensates for inherent non-linearities within image display. Gamma correction is essential for rectifying the inherent non-linear response characteristics of imaging systems used in disease diagnosis, such as monitors and cameras. This process is crucial for producing images that closely resemble reality, by addressing irregularities and ensuring a more accurate representation. In medical imaging, gamma correction is a vital technique for enhancing the visibility of critical diagnostic information. Fig. 2 presents the original and process datasets.

Table 1
Class wise dataset distribution.

Category	Dataset Division			Total Samples
	Train	Validate	Test	
CaS	480	160	160	800
CoS	450	149	149	748
Gum	360	120	120	600
MC	540	180	180	900
OC	324	108	108	540
OLP	540	180	180	900
OT	393	131	131	655

2.3. Pre-trained CNN models

The objective of this research is to establish a robust framework for the diagnosis of oral cancer disease through feature fusion. The proposed approach involves the integration of features from two CNN's constructed using pre-trained models, EfficientNetB0 and EfficientNetB1. The final predictions are generated by fuse the features of these classifiers using a novel self-attention block, following their fusion for classification.

2.3.1. EfficientNetB0

The field of DL has seen remarkable advancements in recent years, particularly in the domain of CNN. These networks have proven highly effective in various computer vision (CV) tasks, ranging from image classification to object detection. However, as models grow in complexity and scale, there is a need to balance accuracy and computational efficiency. EfficientNetB0, introduced by [29], authors address the balance between accuracy and computational efficiency challenge by proposing a novel architecture that optimally scales CNN's depth, width, and resolution. EfficientNetB0 is part of a family of models, denoted by a scaling factor B0, that aims to achieve improved performance while maintaining computational efficiency. The architecture builds upon a combination of neural network design principles, including compound scaling and neural architecture search (NAS). The essential innovation is systematically scaling three dimensions, depth, width, and resolution, by applying a compound scaling factor. This approach allows the network to achieve higher accuracy with fewer parameters and reduced computational complexity. The base architecture of EfficientNetB0 consists of a stack of CNN layers, along with multiple blocks that leverage inverted residual connections. Residual Connections optimize computational resources by reducing parameter count while improving feature representation. Additionally, employing a swish activation function alongside a blend of depthwise and pointwise convolutions enhances the network's efficiency as a whole. Fig. 3 presents the architecture overview of EfficientNetB0 model.

2.3.2. EfficientNetB1

EfficientNet-B1, a member of the EfficientNet CNN family, offers a compelling solution for DL tasks demanding both efficiency and effectiveness. Fig. 4 presents the architecture of EfficientNet-B1. Designed to strike a balance between model size and processing efficiency, EfficientNet-B1 stands as a versatile choice across various CV



Fig. 1. MOD Dataset Sample Images.

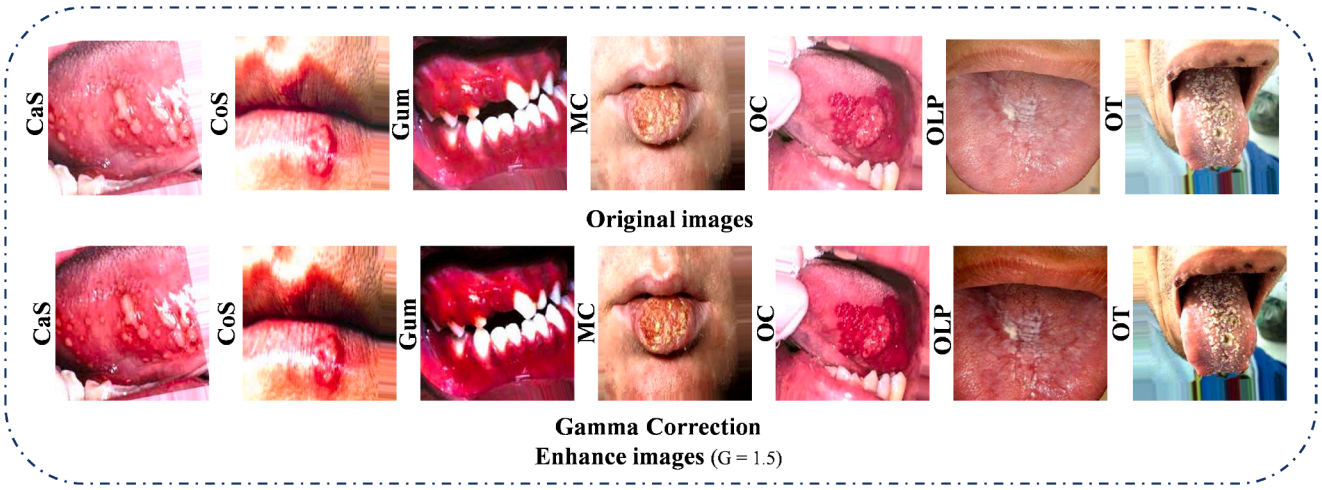


Fig. 2. Comparison between original and enhance gamma correction images.

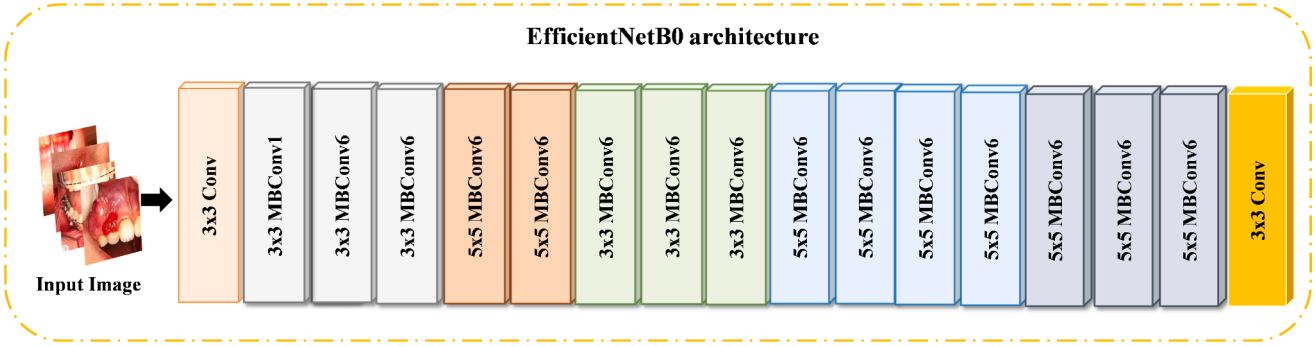


Fig. 3. Architecture overview of the EfficientNetB0 (Base).

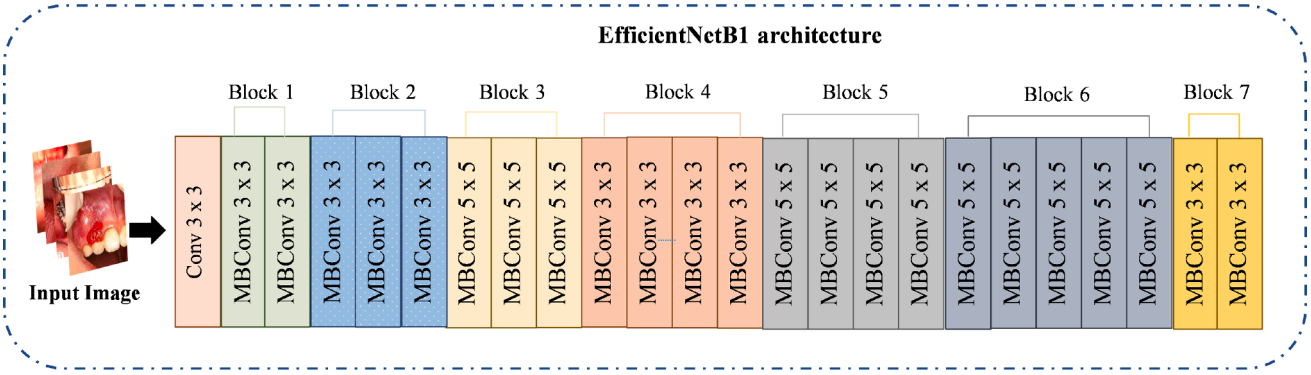


Fig. 4. Architecture overview of the EfficientNetB1 (Base).

applications. This variation deepens and broadens the network architecture at the same time. This is accomplished by [29] introducing a novel compound coefficient (phi), which is logically based on the notion that more complex networks with more layers for more excellent field coverage and more channels for capturing finer details within the bigger image are required for larger input images.

2.3.3. *Feature fusion*

It is suggested that feature-level fusion (Fig. 5) uses two efficientnetB0 and efficientnetB1 models to strike a compromise between model size and processing efficiency. This concept revolves around integrating two efficient models at the feature level. Doing so aims to achieve an equilibrium between model complexity and computational efficiency. This feature-level fusion capitalizes on the unique strengths of each model, combining them to create a unified, efficient, and high-performing solution. This novel approach is poised to excel in scenarios where resource constraints necessitate a careful balance between model size and processing power. This study investigates the feasibility of utilizing the acquired representations of critical features derived from EfficientNet models in the context of oral cancer prediction tasks. A framework has been devised to extract meaningful features by exploiting the capabilities of two outperforming pre-trained models with feature-level fusion.

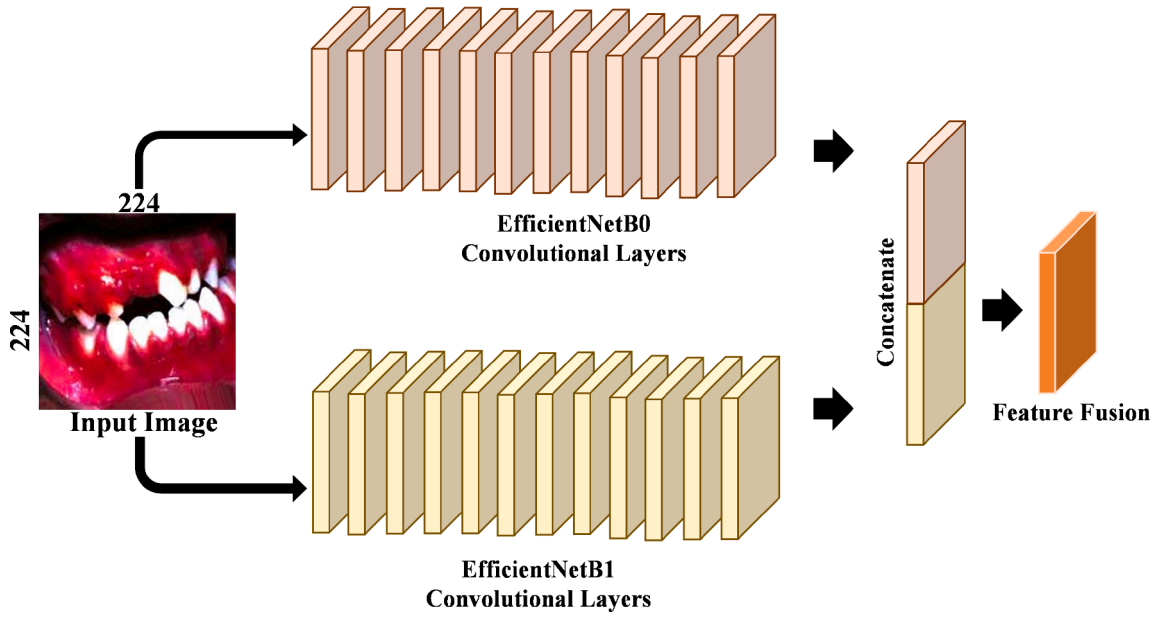


Fig. 5. Overview about Feature Fusion.

2.4. Proposed architecture

Training DL [30] models can be complex due to the inherent randomness and noise present in the training data. These factors can lead to generalization errors and substantial variance. Although regularization and data augmentation are commonly used techniques to mitigate these issues, their effectiveness in DL models can vary. Consequently, more advanced and sophisticated approaches are often necessary to effectively address these challenges. We leveraged the power of EfficientnetB0 and EfficientNetB1 in feature-level fusion, outperforming other pre-trained models for oral cancer prediction. Following the feature fusion, we incorporated the novel self-attention block, introducing an optimal local sparse structure within our CNN. This approach enables us to utilize multiple filter sizes within a single image block rather than being confined to a single filter size. The outputs are concatenated and passed to the next layer. This innovative method enhances the reliability and accuracy of predictions by combining feature fusion with the novel self-attention block. Through this fusion-based attention learning, we effectively reduce prediction error variance and surpass the performance of individual contributing models. Our proposed model significantly elevates performance in classifying oral cancer images.

Our main objective in creating the feature fusion architecture was to surpass the capabilities of existing state-of-the-art techniques, ultimately leading to accurate oral cancer disease diagnosis and patient categorization. We employed a feature fusion attention technique known for its potent fusion capabilities, capable of markedly enhancing performance to achieve this. By incorporating the novel self-attention block, we seamlessly integrate the selected top-performing base models to construct our proposed architecture during the feature fusion phase, as depicted in Fig. 6. Algorithm 1 provides a more comprehensive insight into the entire suggested approach.

Algorithm 1: Proposed Architecture for Mouth and Oral Disease Detection

/ **Preprocessing Phase** */*
Input: *original MOC Dataset*
Gamma correction value G:1.5; *the enhance the image discernibility*
Output: *enhance the visibility of critical image diagnostic & predict oral cancer*
/ **Feature Fusion Architecture** */*

(*continued on next page*)

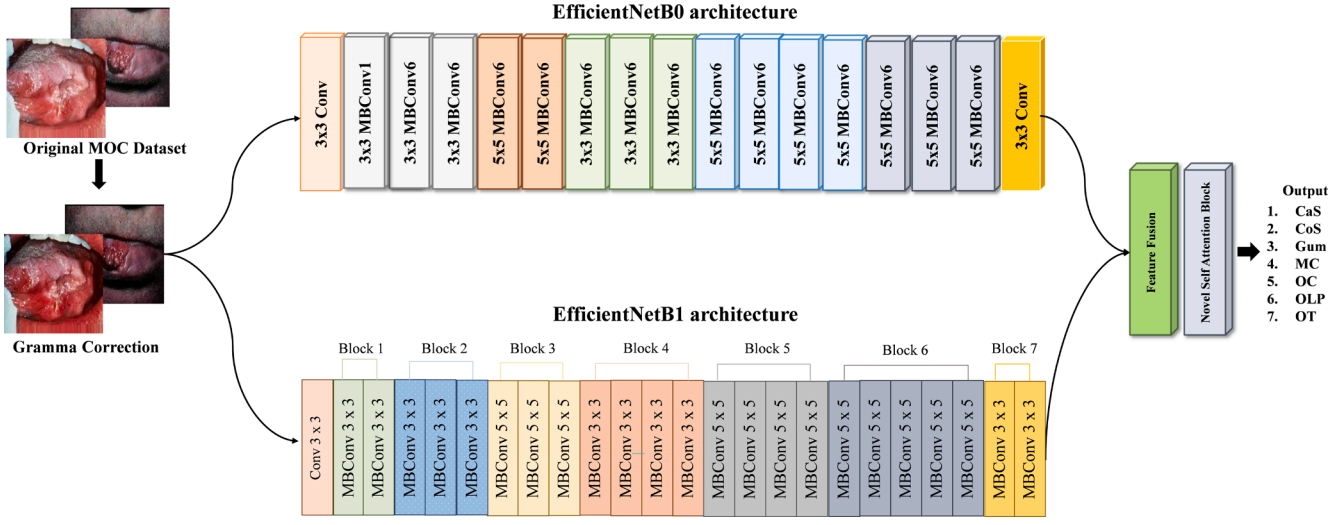


Fig. 6. Feature Fusion Architecture; which fuse features of two EfficientNet models through novel self-attention to prediction cancer category.

(*continued*)

	Algorithm 1: Proposed Architecture for Mouth and Oral Disease Detection
1	*Load gamma corrected enhance images*
2	*Import the two models*; *EfficientNetB0, and EfficientNetB1*
3	*Load train weights*
4	*Features fusing using concatenation layer*
5	*Integration of novel self-attention block*
6	*Test dataset load*;
	Output the prediction of all test dataset; *Evaluate Feature Fusion Architecture Score*; *Classification Report*; *Confusion Matrix*; *Ablation Study*;

2.4.1. Novel self attention block to optimize fused feature

We have introduced a novel self-attention block to enhance feature fusion within the EfficientNet model. In this novel block, after feature fusion Dense layer is applied to the fused feature with 128 neurons and a ReLU activation function. This dense layer learns to assign weights to different parts of the fused feature, effectively capturing the importance of each feature. The output of the dense layer is then flattened using the Flatten layer, transforming it into a one-dimensional vector. To ensure that the computed weights sum up to 1 (as attention weights should), the softmax activation function is applied to the flattened vector. This softmax operation normalizes the values to form a probability distribution, with each value representing the attention weight of the corresponding input feature. To match the shape of the original input feature, the attention weights need to be reshaped appropriately. Repeat-Vector layer is used to replicate the attention weights along the last dimension. This ensures that the attention weights have the same length as the input feature. The Permute layer is then applied to swap the dimensions of the attention weights, making them compatible for element-wise multiplication with the input feature. Multiply layer is employed to perform element-wise multiplication between the input feature and the reshaped attention weights. This operation effectively "attends" to different parts of the input feature based on the computed attention weights. The result of this multiplication is the attended output attention weights, which highlights feature in the feature fusion that are most relevant according to the learned attention weights. Fig. 7 display the architecture overview of the novel self-attention block.

3 Novel self-attention block mathematical formulation for enhancing feature fusion for oral cancer prediction

Here Y denote the input shape (m,n) of image, where m is the number of instance and n is the number of features.

4 Step 1: Compute attention weights

- First, apply a dense layer to the input Y, resulting in $W_a{*}Y + b_a$, where W_a represents the weights matrix and b_a represents the bias vector of the dense layer. This operation is repented as $Z = ReLU(W_a{*}Y + b_a)$.
- Next, flatten the output Z to obtain a vector representation.
- Apply the *Softmax* activation function to obtain attention weights, denoted as a. This operation is represented as $a = Softmax(flatten(Z))$.

5 Step 2: Modify attention weights

- Repeat the attention weights n times (to match the dimensions of the input (Y)) using *RepeatVector*, obtaining an input shape (m, n, c).
- Permute the dimensions of the repeated attentions weights to match the shape of the input Y, resulting in an input shape (m, n, c).

Step 3: Apply attention weights to the input tensor

$\hat{Y} = Y \odot a$, where $\odot$ represents elements-wise multiplication.

The final output of the attention block, $\hat{Y}$ obtained from the element-wise multiplication.

3. Results and discussion

This section begins by outlining the experimental configuration employed in the present study. After that, delve into the specifics of hyper parameters, the performance metrics used for model evaluation, and an analysis of feature maps. Following that, showcase the outcomes of the proposed method and draw comparisons with established approaches to highlight its superior performance.

3.1. Experimental setting

The study's Feature Fusion model was trained using the Keras framework and Python programming. All experiments were conducted within the widely-used Python environment to exploits the GPU runtime type and computational capabilities. Model testing and training were performed using 8 MB of RAM and an NVIDIA GeForce MX350 Tesla GPU.

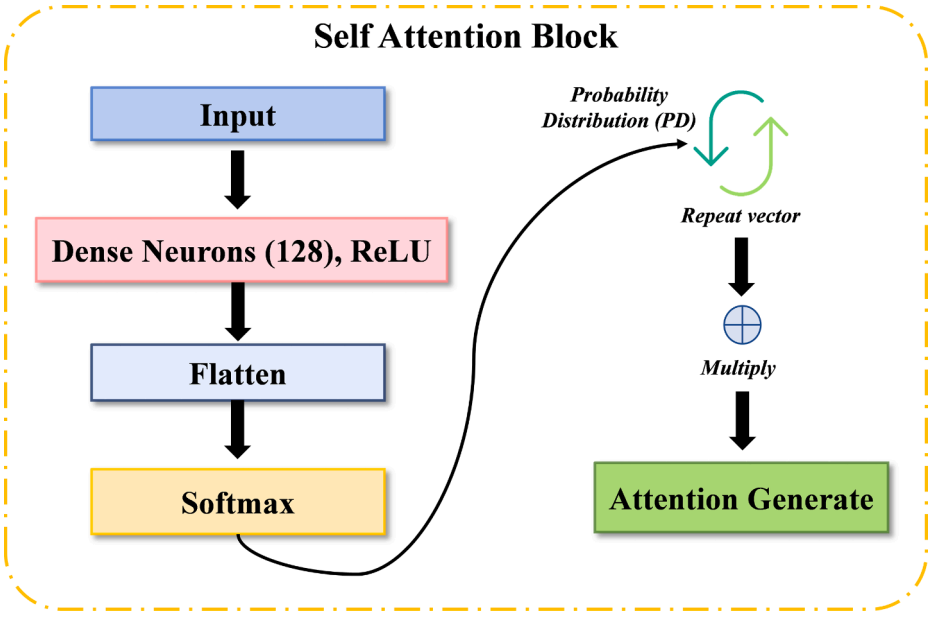


Fig. 7. Architecture overview of the novel self-attention block.

3.2. Model hyper parameter

We ensured the suggested model's consistency by training it with the best hyper parameters. Specifically, our training procedure comprised 64 batch sizes, taught over 10 epochs, and employed a learning rate of 0.001. We have noteworthy improvements in the model's performance to the efficient utilization of the Adam optimizer. The Adam optimizer exhibited superior accuracy and reduced memory usage compared to alternative optimizers. Our training process depends on utilizing the categorical cross-entropy loss function, underscoring its pivotal role in our methodology for effectively evaluating and enhancing overall performance.

3.3. Performance metrics

We utilized well-established evaluation criteria commonly employed in DL tasks to assess the efficacy of our proposed feature fusion architecture, and our findings align closely with prior research outcomes. For each diagnosed sample, we categorized the diagnosis as True Positive-TP, True Negative-TN, False Positive-FP, or False Negative-FN. TP and TN represent correct classifications, while FP and FN indicate incorrect diagnoses. The mathematical expressions below outline the formulas for these performance metrics:

$$Accuracy(AC) = \frac{TP + TN}{TP + FP + TN + FN} \tag{1}$$

$$Precision(PR) = \frac{TP}{TP + FP} \tag{2}$$

$$Recall(RC) = \frac{TP}{TP + FN} \tag{3}$$

$$F1Score(FS) = 2 \times \frac{PrecionxRecall}{Precision + Recall} \tag{4}$$

3.4. Classification results

The Table 2 provides performance metrics for a classification model, specifically detailing precision, recall, F1-score, and accuracy for different classes. In the *CoS* class, the model achieved a high precision of 1.00, indicating that all the instances it classified as *CoS* were indeed *CoS*. The recall is 0.99, implying it correctly identified 99 % of all *CaS* instances. The F1-score is also a perfect 0.99, suggesting enhanced overall performance. The model's accuracy for overall is 98.83 %, showcasing its effectiveness in correctly classifying each class.

Similar near-perfect performance is observed in classes *Gum, OLP*, and *OT*, with precision, recall, F1-score, and accuracy all being very close to 1.00. These results indicate that the model is highly accurate in classifying these categories. However, for classes *MC* and *OC*, while the precision and recall are still relatively high at 0.98, respectively, the F1-score is slightly lower at 0.99. This suggests that the model's performance may be less balanced for these classes, but it still maintains a high accuracy overall.

The confusion matrix depicted in Fig. 8 illustrate the predicted outcomes of the proposed feature fusion architecture. Out of the 1028 images within the test set, the suggested model demonstrated impressive performance, accurately classifying all images across various classes. It encountered only four misclassifications, which can be specified as follows: one image was mistakenly assigned to the MC class, one image to the OC class, one images to the OT class, and one image to the CaS class.

Table 2
Class-wise performance of proposed model.

Class	Precision	Recall	F1-Sore	Accuracy
CaS	0.99	0.99	0.99	***98.83***
CoS	1.00	1.00	1.00	
Gum	1.00	1.00	1.00	
MC	0.98	0.98	0.98	
OC	0.98	0.98	0.98	
OLP	0.99	0.99	0.98	
OT	0.99	0.98	0.98	

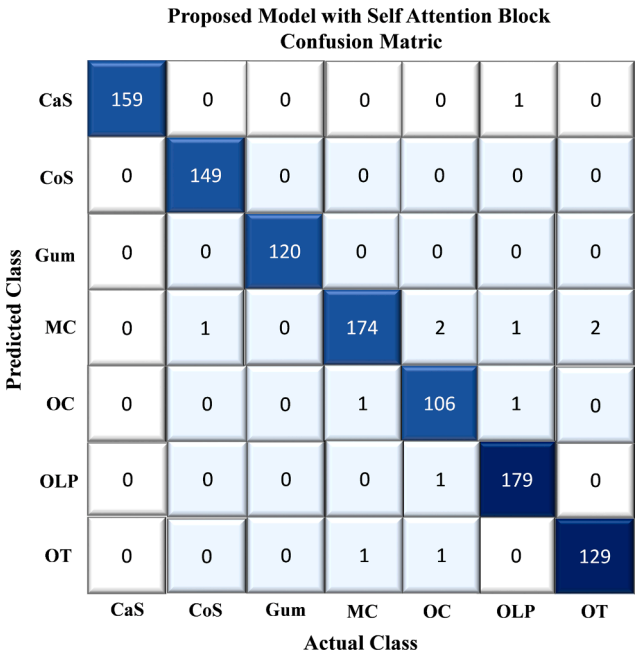


Fig. 8. Confusion Matric of Feature Fusion Model.

3.5. Feature map analysis

Visual analysis of deep models is crucial in practical applications, as it significantly enhances interpretability. In this context, we explore how our proposed baseline model employs visible and interpretable insights from the EfficientNet model to diagnose specific regions as features. To better understand the EfficientNet model's functioning, we present a feature map analysis applied to oral cancer images in Fig. 9. Upon visual examination, it becomes apparent that our model offers enhanced interpretability and precise identification of the affected areas necessary for categorization. Furthermore, in Fig. 9, we showcase intermediate feature maps from various convolution blocks of the EfficientNet model. These feature maps allow us to observe how the model divides its input into distinct filters, elucidating how it extracts multiple features from the same input image.

3.6. Performance comparison with other blocks

To validate the effectiveness of proposed novel block. Table 3 presents a comparison of various neural network block architectures, including Residual Block, Naïve Inception Block, Dense Block, with a Novel Self-Attention Block. Residual block Achieves high precision, recall, f1-score, and accuracy, indicating strong performance in classification tasks. However, one disadvantage of the Residual Block is its tendency to suffer from vanishing gradients, especially in very deep networks, which can delay training convergence. Naïve Inception Block shows competitive performance across all metrics, though slightly lower than the Residual Block and Dense Block. A disadvantage of the Naïve Inception Block is its increased computational complexity due to the parallel operations within the block. Dense Block shows performs consistently well with high precision, recall, f1-score, and accuracy. However, Dense Blocks can lead to significantly increased model size and computational cost, particularly in deeper networks, which can pose challenges in deployment and inference. Novel Self-Attention Block

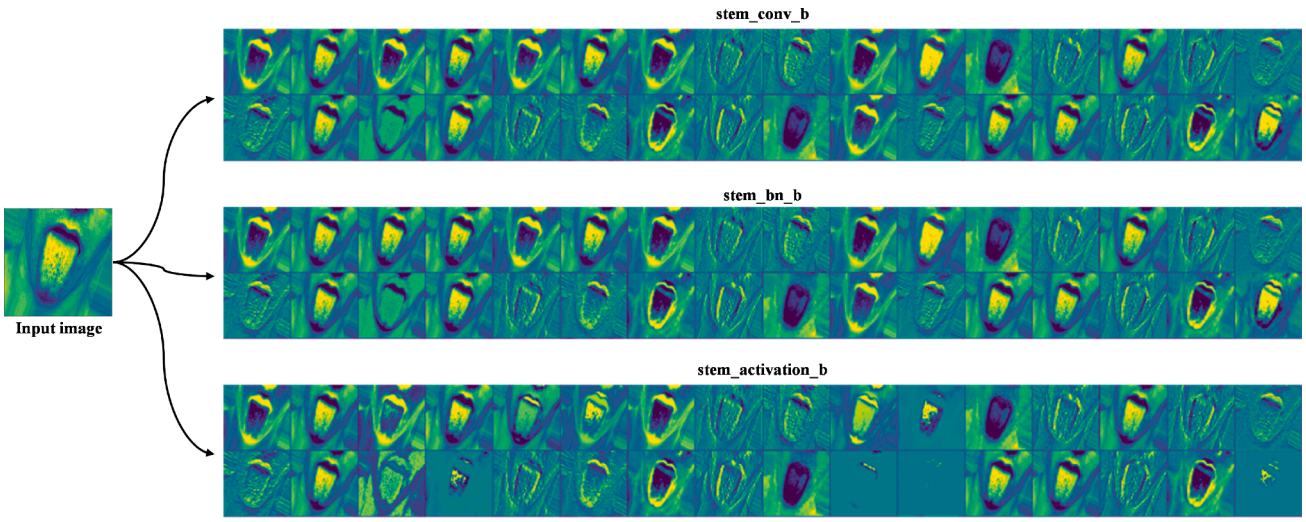


Fig. 9. Feature Map Analysis of Baseline EfficientNet Model.

Table 3
Performance comparison of proposed novel self-attention block with existing blocks.

Models	Precision	Recall	F1-Score	Accuracy
Residual Block	95.66	96.21	95.86	95.91
Naïve Inception Block	95.26	95.45	95.31	95.33
Dense Block	95.66	95.71	95.66	95.71
Novel self-attention Block	**98.73**	**98.82**	**98.82**	**98.83**

demonstrates exceptional performance across all metrics, outperforming other blocks. Additionally, the proposed Self-Attention Block efficiently combines dense layers, repeated vectors, and multiplication layers, which enhances its capability to capture complex dependencies across input sequences, leading to improved performance and generalization.

3.7. Performance analysis with existing models

In the Table 4 provided, two different models' accuracy scores are compared: InceptionResNetV2 [27] model achieved an accuracy of 99.50 %, while the proposed model, which utilizes a Feature Fusion self-attention approach, achieved a slightly lower accuracy of 98.83 %. Although the difference in accuracy between these models is minimal, it's worth noting that the proposed model introduces a significant improvement in terms of result explainability when compared to the previous model presented [27]. Unlike the previous model, the proposed model strongly emphasizes providing explainable results. This means that it achieves a high level of accuracy and offers insights and explanations for how and why specific results were obtained. This focus on result explainability is a crucial step towards making the model's predictions more transparent and understandable, which is especially important in fields where model interpretability is essential for decision-making and trust.

3.8. Pre-trained backbone models comparison

The Table 5 compares different DL models' performance metrics on a specific task, likely a classification task. These models include ResNet-50, ResNet-101, VGG-16, VGG-19, EfficientNet-B4, EfficientNet-B5, EfficientNet-B7, and a Proposed self-attention Model. The performance metrics evaluated are Precision, Recall, F1-Score, and Accuracy. Among the models, the Proposed Model outperforms the others in Precision, Recall, F1-Score, and Accuracy, achieving exceptionally high scores of approximately 98.73 %, 98.82 %, 98.77 %, and 98.83 %, respectively. This indicates that the Proposed Model exhibits high accuracy and reliability in correctly classifying the data it tested on. EfficientNet-B4 also performs well with high scores across all metrics, while VGG-16 and VGG-19 lag in performance, with lower Precision, Recall, F1-Score, and Accuracy values. ResNet-50 and ResNet-101 perform well but are slightly lower than EfficientNet-B4 and the Proposed Model. Overall, the Proposed Model is the top-performing model in this comparison.

Table 4
Performance analysis of proposed model with existing models.

Reference	Method	Accuracy
Rashid et al [27]	InceptionResNetV2	99.50
Proposed Model	**Feature Fusion self-attention Approach**	**98.83**

Table 5
Classification comparison of the proposed feature fusion model with advance TL models.

Models	Precision	Recall	F1-Score	Accuracy
ResNet-50 (Base)	92.67	91.07	91.68	91.63
ResNet-101 (Base)	92.08	91.44	91.69	91.62
VGG-16 (Base)	82.70	81.60	82.06	81.90
VGG-19 (Base)	81.48	82.49	81.64	81.42
EfficientNet-B4 (Base)	94.18	94.23	94.10	94.16
EfficientNet-B5 (Base)	92.67	93.03	92.75	92.70
EfficientNet-B7 (Base)	92.38	92.89	92.54	92.70
Proposed Model	**98.73**	**98.82**	**98.82**	**98.83**

3.9. Ablation study

Table 6 presents a comparison of baseline models based on several performance metrics for a specific task, likely in the field of DL or data analysis. Four models are evaluated: EfficientNet-B0 (baseline),

Table 6
Demonstrate the performance of the proposed Feature Fusion architecture, including backbone model, feature fusion without novel block, and with novel block.

Models	Precision	Recall	F1-Score	Accuracy
EfficientNet-B0	94.47	94.55	94.46	94.06
EfficientNet-B1	95.20	94.99	95.06	95.03
Proposed WO Block	98.17	98.41	98.28	98.34
Proposed Model	**98.73**	**98.82**	**98.77**	**98.83**

EfficientNet-B1(baseline), Proposed WO Block, and Proposed Model. These models are assessed in precision, recall, F1-Score, and accuracy. Among them, the Proposed Model stands out with the highest scores across all metrics, demonstrating excellent performance with a precision of **98.73** %, recall of **98.82** %, F1-Score of **98.82** %, and an accuracy rate of **98.83** %. This suggests that the Proposed Model is highly effective in the given task, surpassing the other models in terms of overall performance.

Fig. 10 presents a detailed analysis comparing the actual and predicted values for our proposed feature fusion architecture under various conditions. Specifically, we examine the model's performance with and without the novel self-attention block. Our study involved a test set comprising 1028 images, and the results are quite promising. The suggested model consistently showcased remarkable performance, successfully classifying 1016 images across various classes. This demonstrates the robustness and efficacy of our proposed feature fusion approach.

However, it's worth noting that the model's performance experienced a notable decline when the novel self-attention block was omitted. This indicates the crucial role played by the novel self-attention block in enhancing the model's predictive capabilities. Its absence resulted in a decrease in overall accuracy and the ability to generalize across different classes of images.

Furthermore, we compared the performance of our proposed model to a baseline model, which was trained separately. Surprisingly, the baseline model also exhibited strong predictive abilities, outperforming other pre-trained models. This observation suggests that our proposed feature fusion architecture, when integrated with the novel self-attention block, matches and surpasses the performance of well-established models. Our study demonstrates a step-by-step

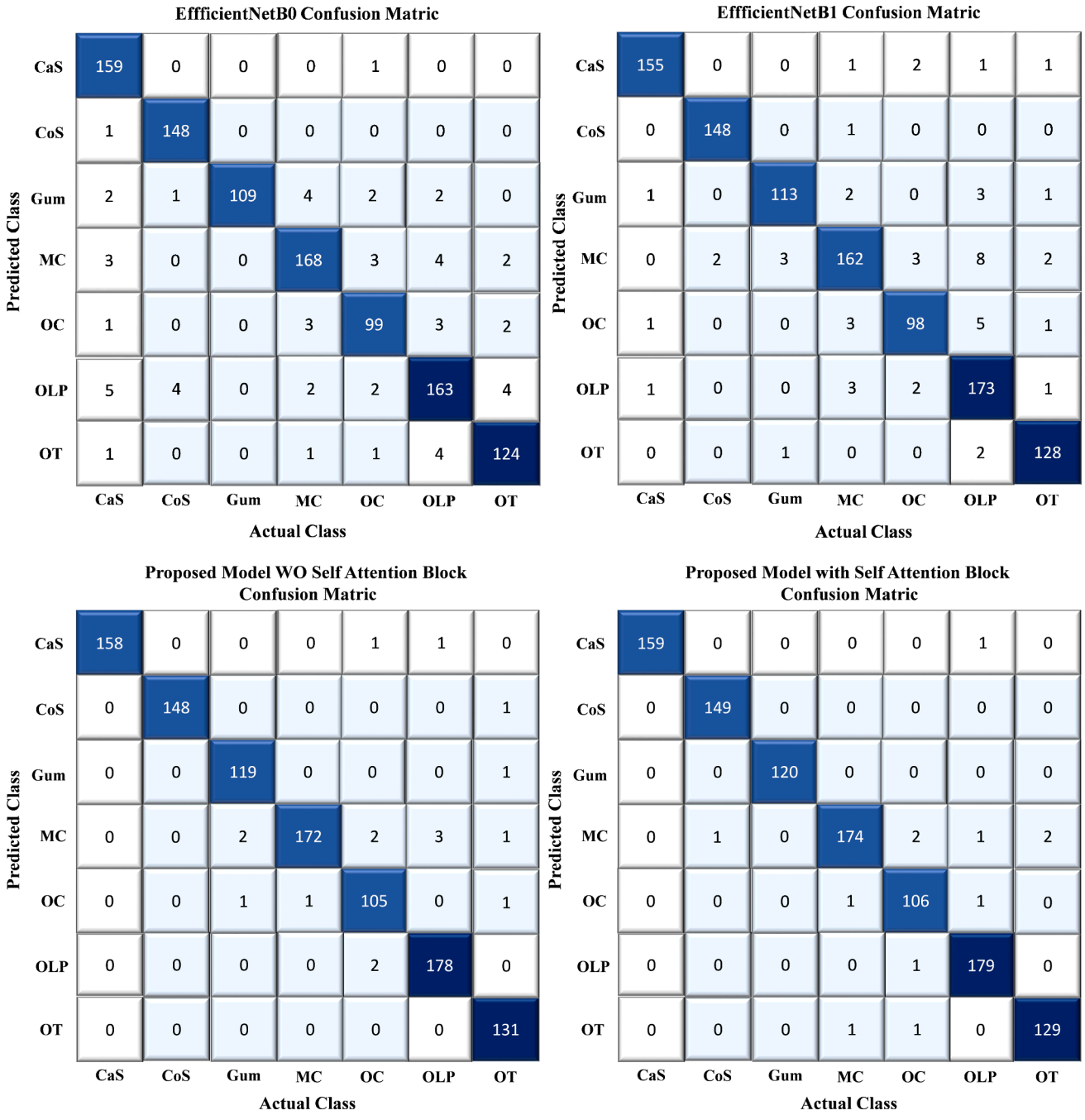


Fig. 10. Confusion Analysis Baselines, Proposed model with or without novel self-attention Block.

improvement in the model's performance. Starting with the baseline, adding the novel self-attention block and our feature fusion approach consistently enhanced the model's accuracy and robustness. These findings underscore the effectiveness of our methodology and highlight the potential for further advancements in image classification tasks.

The results in Fig. 11 compare the Kappa coefficient and error values of several models on the test set. After analyzing the tested base models, which included the model with novel self-attention block, EfficientNet-B0 (Baseline), and EfficientNet-B1 (Baseline), kappa coefficients of 0.936, 0.941, and 0.980 were discovered. By comparison, the suggested feature fusion strategy significantly improved the Kappa value, with a coefficient of 0.983. Furthermore, the graph shows that the model's error value drops with increasing Kappa coefficient, suggesting that the suggested model becomes more accurate at making predictions as Kappa values rise. Given its significant improvement in the Kappa coefficient, the proposed model appears to have a high level of predictability and resilience. The findings show that the suggested model has improved the model's performance. The observed gains in the f1-score values, Kappa coefficients, and accuracy scores of the proposed model over the tested base models serve as evidence.

4. Limitation and future work

In the early stages of our research on oral cancer, we encountered certain limitations that fell beyond the scope of our study. As a result, we acknowledge the following constraints inherent in our work, which could potentially be addressed in future investigations by adopting appropriate tools and methodologies:

- The MOD dataset is the sole publicly accessible resource for oral images. The limited availability of reference materials and datasets poses a significant challenge in accurately evaluating our models' efficacy. This issue is particularly acute in the context of oral cancer, as the condition impacts the entire oral cavity, making it challenging to procure images of affected faces. There is a pressing need for a more extensive dataset to enhance the clinical testing process for healthcare professionals. Fortunately, several public repositories diligently collect fresh data, augmenting the available resources. Consequently, the absence of data may soon be alleviated, opening doors for future studies to comprehensively assess the performance of proposed models.
- Recreating models using patient datasets that encompass symptoms, medical histories, and age profiles can yield more authentic and precise results. Nevertheless, procuring patient data for research purposes can pose significant challenges. Our aspiration is that hospitals will, over time, facilitate the availability of such data to boost the efforts of researchers in their quest for improved medical insights.
- We can enable real-time patient diagnosis in resource-constrained regions by deploying our proposed model within a mobile application, potentially leading to life-saving outcomes. Further exploration of this potential will be a focal point in our upcoming research endeavors.

5. Conclusion

Oral cancer is a disease with far-reaching implications for quality of life, prompting global efforts from researchers to uncover effective treatments, implement strategies for containment, and mitigate its impact on public health. In our research of innovative ways to predict the progression of oral cancer, we have developed a novel approach known as Feature Attention-Level Fusion. This method exploits the power of novel self-attention block integration to amalgamate the distinctive features of two foundational models, EfficientNet-B0 and EfficientNet-B1. Notably, this makes the first instance of self-attention module fusion at the feature level, leveraging the attributes of EfficientNet-B0 and EfficientNet-B1 and employing various convolutional filter sizes to capture features at multiple scales within the input data. It is worth highlighting that no previous research has explored this initial feature-level fusion technique designed to enhance the allocation of weights. Through an evaluation using the sole publicly available MOD dataset, we ascertained that our proposed model outperforms TL models and previous methodologies, showcasing 98.83 % accuracy in oral cancer prediction. Previous research has not explored utilizing the initial, straightforward feature-level fusion technique to enhance feature weight allocation. In our study, leveraging the sole publicly available

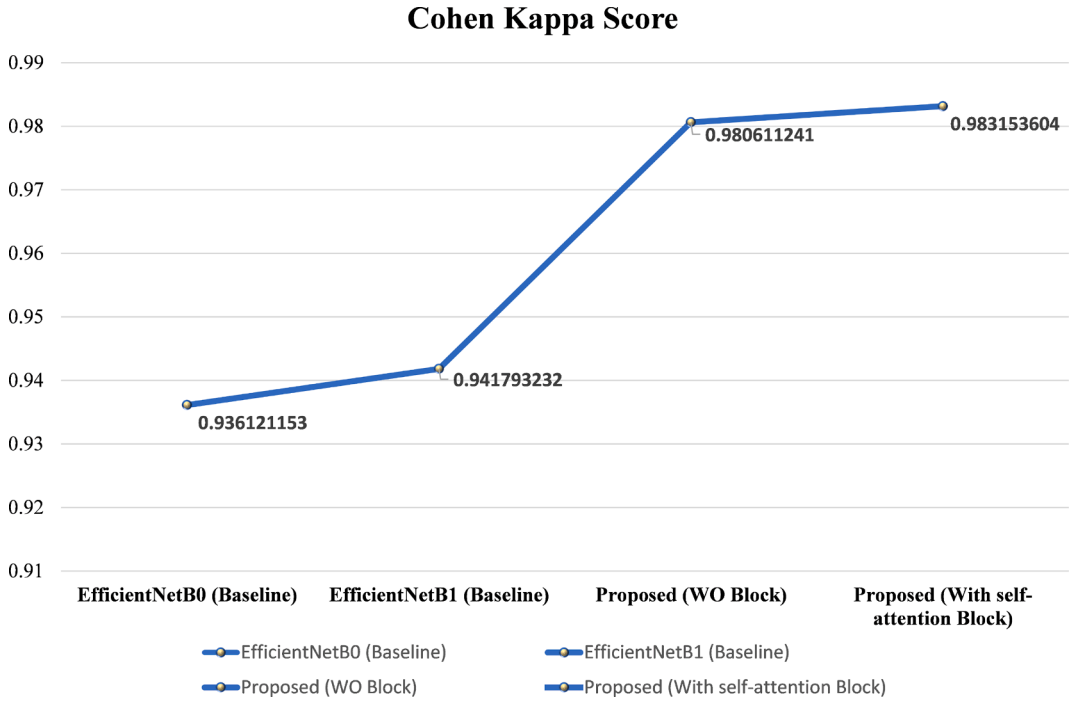


Fig. 11. Cohen's Kappa Coefficient compares the suggested model's reliability to the base models.

MOD dataset, we evaluated the proposed model's performance and observed that it outperformed prior methodologies.

We aim to explore various ensemble methods to enhance performance in the future. Additionally, as outlined in Section 4, we plan to augment our dataset with more samples and develop a real-time mobile application. These initiatives are expected to help us overcome some of the current limitations we face.

CRediT authorship contribution statement

Saif Ur Rehman Khan: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Funding acquisition, Conceptualization, Data curation. **Sohaib Asif:** Writing – review & editing, Visualization, Validation, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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